Lecture 16

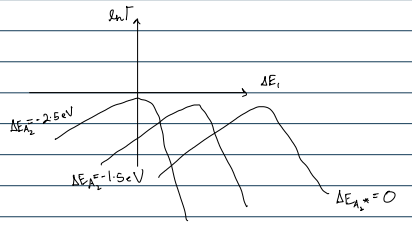
Tuesday, 21 March 2023 10:32 AM

Recall:

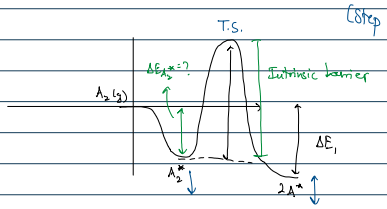
1) $A_2(g) + * \rightleftharpoons A_2^*$

2) $A_2^* + * \rightleftharpoons 2A^*$

3) $A^* + B(g) \rightleftharpoons AB(g) + *$



(Step 2 is RLS)



Moving A_2^* will increase the E for TS → So ΔE_1 has to move proportionally down to keep the same barrier

What if there is more than 1 rate controlling step?

1) $A_2(g) + 2* \rightleftharpoons 2A^*$

2) $A^* + B(g) \rightleftharpoons AB(g) + *$

A_2^* → corresponds to the binding energy → If A_2^* goes really high up then the RLS will change

$$r_1 = k_1 \theta_*^2 P_{A_2} - k_{-1} \theta_A^2$$

$$r_2 = k_2 \theta_A P_B - k_{-2} \theta_* P_{AB}$$

$$r_1 = k_1 \theta_*^2 P_{A_2}(1-\gamma_1) \quad ; \quad \gamma_1 = \frac{\theta_A^2}{\theta_*^2 P_{A_2}' K_1}$$

$$r_2 = k_2 \theta_A P_B(1-\gamma_2) \quad ; \quad \gamma_2 = \frac{\theta_* P_{AB}'}{\theta_A P_B' K_2}$$

$$\gamma_{overall} = \gamma_1 \gamma_2^2$$

Steady state for A^*: $2r_1 = r_2$

Site balance: $\theta_* + \theta_A = 1$

$$2k_1 P_{A_2} \overbrace{(1-\theta_A)}^{\theta_*})^2 - 2k_{-1}\theta_A^2 = k_2 \theta_A P_B - k_{-2} P_{AB} \overbrace{(1-\theta_A)}^{\theta_*}$$

(quadratic in θ_A)

$$\theta_A = \frac{1 + \frac{k_{-2} P_{AB}}{2k_1 P_{A_2}} \left(1 + K_2 \frac{P_B}{P_{AB}}\right)}{1 - \frac{1}{K_1 P_{A_2}'}} \left[1 - \sqrt{1 - \left(1 - \frac{1}{K_1 P_{A_2}}\right) \frac{\left(1 + \frac{k_{-2} P_{AB}}{2k_1 P_{A_2}}\right)}{\left(1 + \frac{k_{-2} P_{AB}}{2k_1 P_{A_2}} \left(1 + K_2 \frac{P_B}{P_{AB}}\right)\right)^2}} \right]$$

$$r = r_1 = k_1 P_{A_2}(1-\theta_A)^2 - k_{-1}\theta_A^2$$

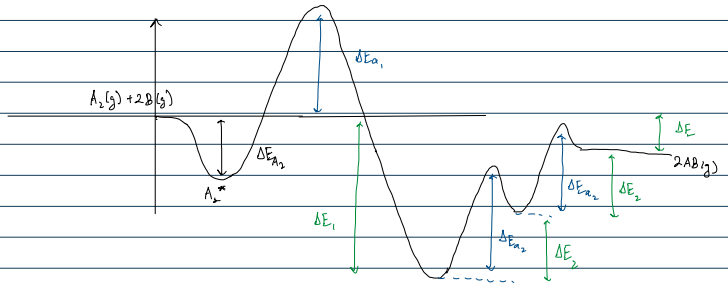
$$= k_1 P_{A_2}(1-\theta_A)^2 \left[1 - \frac{\frac{\theta_A^2}{(1-\theta_A)^2 P_{A_2}'}}{K_1} \right]$$

Exercise: Show that θ_A reduces to QE exp when

a) $\frac{k_{-2}}{k_1} \ll 1$ b) $\frac{k_{-2}}{k_1} \gg 1$

Step 2 is RLS Step 1 is RLS

Overall Energy Diagram



$$k_1 = k_{1_0} e^{-\frac{\Delta E_{a_1}}{k_B T}} \quad ; \quad k_{1_0} = \frac{k_B T}{h P_0} e^{\frac{\Delta S_1^{o\ddagger}}{k_B}} \quad ; \quad \Delta S_1^{o\ddagger} = -200 \frac{J}{mol\,K}$$

$$K_1 = e^{\Delta S_1^\circ / k_B} e^{-\Delta E_1 / k_B T}$$



$$\Delta E_{a_1} = \alpha_1 \Delta E_1 + \beta_1 \quad ; \quad K_2 = e^{\Delta S_2^\circ / k_B} e^{-\Delta E_2 / k_B T}$$

$$k_2 = \frac{k_B T}{h P^\circ} e^{\Delta S_2^{o\ddagger} / k_B} e^{-\Delta E_{a_2} / k_B T} \quad ; \quad k_{-2} = \frac{k_2}{K_2} \quad ; \quad \Delta S_2^{o\ddagger} \approx -200 \frac{J}{mol\,K}$$

$$\Delta E_{a_2} \approx \alpha_2 \Delta E_2 + \beta_2 \qquad \alpha_2 < 0$$

$$\alpha_2 \approx -\frac{1}{2} \quad ; \quad \beta_2 \approx 0.8\ eV$$

} Values can vary in the literature

where does this come from? physical meaning?

$$\Delta E_2 = \frac{\Delta E_{ov} - \Delta E_1}{2} \quad ; \quad \text{choose } \Delta E_{ov} \approx -1\ eV$$

$$\Delta S_2^\circ \propto S_{AB(g)}^\circ - S_{B(g)}^\circ \approx 0 \quad ; \quad \gamma = \frac{(P_{AB}/P_B^\circ)^2}{k_1 k_2^2 P_{A_2}'} \qquad \text{Assume, } P_{A_2}' = 25$$

↑ Standard entropy change for step 2

$$\Downarrow$$

$$\frac{P_{AB}}{P_B} = \sqrt{k_1 k_2 P_{A_2}' \gamma} \quad ; \quad P_B' \approx 50$$

↑ treat as a parameter (or add a reactor model)

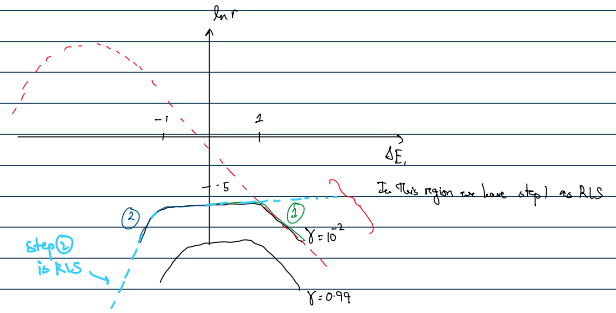
→ design equations for PFR, PBR

} Can solve for microkinetic eq & reactor eq to solve for partial pressures across the reactor

Note: ① Values of $S_s^{o\prime}$, P_s', γ have a big effect on rate, but a smaller effect on volcano maximum

② BEP relationships affect everything!

Volcano: Assuming that first step is a lumped step; → Don't need binding/activation energy of step 2!



Different "legs" of volcano (① & ③) correspond (approx.) to different assumptions about RLS's

Leg ①: Assume step 1 is RLS (as before)

$$r = r_1 = k_1 P_{A_2} \theta_*^2 - k_{-1} \theta_A^2 \quad ; \quad \theta_A = k_2^{-1} \frac{P_{AB}}{P_B^\circ} \theta_*$$

} γ → overall approach to equilibrium

Leg ①: Assume step 1 is RLS (as before)

$r = r_1 = k_1 P_{A_1} \theta_\alpha^2 - k_{-1} \theta_A^2$; $\theta_A = K_1^{-1} \frac{P_{A_2}}{P_B} \theta_\alpha$

$\gamma \rightarrow$ overall approach to equilibrium

$r = k_1 P_{A_2} \theta_\alpha^2 (1-\gamma)$

$r_① = \frac{k_1 P_{A_2}}{\left(1 + K_1^{-1} \frac{P_{A_2}}{P_B}\right)^2} (1-\gamma)$; Sabatier: assume max rates $\rightarrow \theta_\alpha = 1$

$r_{cat} \approx k_1 P_{A_2} (1-\gamma)$ (upper bound)

↓ Sketch on above graph

$r_②$: assume step 2 is RLS:

$r = \frac{1}{2} r_② = \frac{1}{2} (k_2 P_B \theta_A - k_{-2} P_{A_2} \theta_\alpha)$

Step 1 is QE: $\theta_A = \sqrt{K_1 P_{A_2}'}\, \theta_\alpha$; $\theta_A + \theta_\alpha = 1$

$r_② = \frac{1}{2} k_2 P_B \theta_A (1 - \sqrt{\gamma})$

$r_③ = \frac{\frac{1}{2} k_2 \sqrt{K_1 P_{A_2}'}}{1 + \sqrt{K_1 P_A^2}} (1 - \sqrt{\gamma})$ < upper bound $\frac{1}{2} k_2 P_B \ldots$

Note: 1) The legs (sides) are determined by assuming different RLS's. The max of the volcano is a transition between different RLS

[Each RLS has its own volcano - competition b/w rate constants and coverage effects]

2) The limiting expressions approximate the true volcano except where volcanos cross

3) Sabatier rates are useful if $\gamma \ll 1$

4) Volcanos assuming only 1 RLS can wildly overestimate rates

↳ Volcanoes w/ > 2 steps may have many legs!